

LAPORAN PRAKTIKUM ASD



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Sistem Informasi Bisnis / 1E

JOBSHEET 6 Sorting (Bubble, Selection, Insertion Sort)

5.1 Tujuan Praktikum

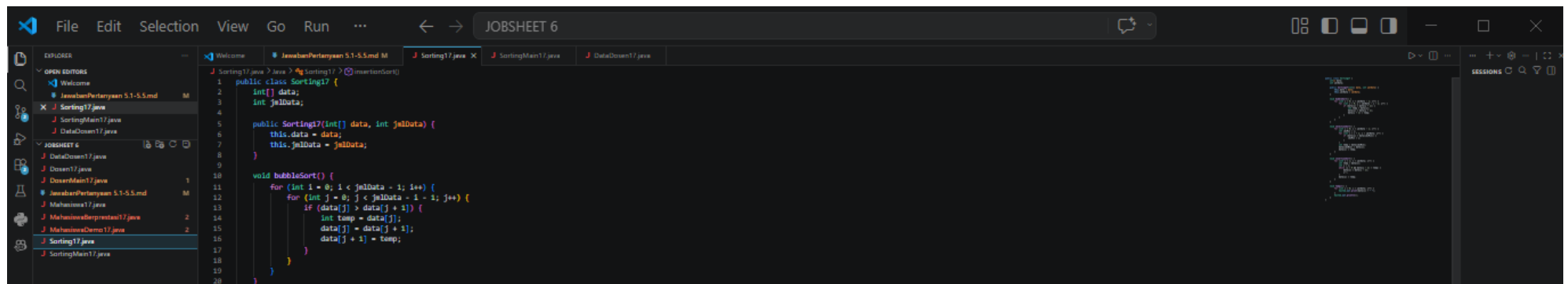
1. Memahami algoritma sorting (Bubble, Selection, Insertion)
2. Mampu mengimplementasikan sorting dalam program Java

5.1 Tujuan Praktikum

1. Memahami algoritma sorting (Bubble, Selection, Insertion)
2. Mampu mengimplementasikan sorting dalam program Java

5.2 Praktikum 1

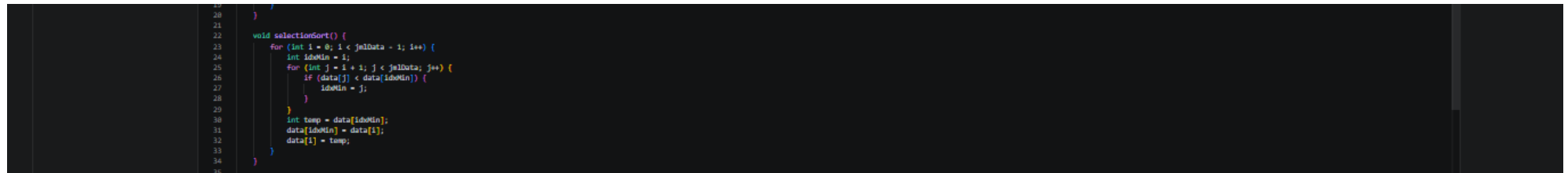
A. Bubble Sort



The screenshot shows an IDE with the following code in `Sorting17.java`:

```
1 public class Sorting17 {  
2     int[] data;  
3     int jmlData;  
4  
5     public Sorting17(int[] data, int jmlData) {  
6         this.data = data;  
7         this.jmlData = jmlData;  
8     }  
9  
10    void bubbleSort() {  
11        for (int i = 0; i < jmlData - 1; i++) {  
12            for (int j = 0; j < jmlData - i - 1; j++) {  
13                if (data[j] > data[j + 1]) {  
14                    int temp = data[j];  
15                    data[j] = data[j + 1];  
16                    data[j + 1] = temp;  
17                }  
18            }  
19        }  
20    }
```

B. Selection Sort



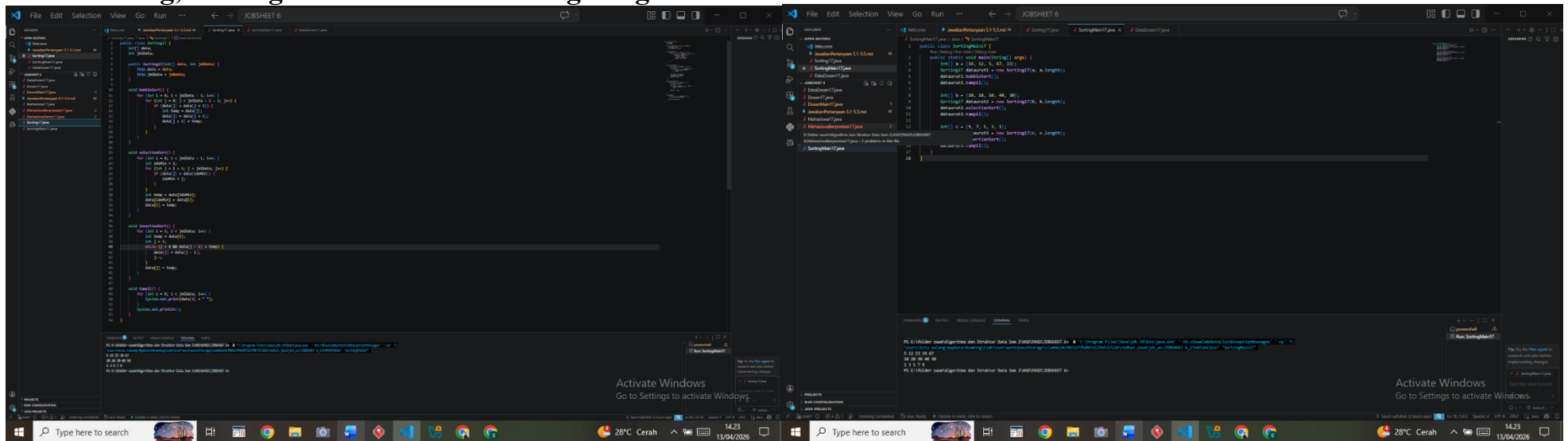
The screenshot shows the Selection Sort implementation in the `void selectionSort()` method of `Sorting17.java`:

```
21  
22    void selectionSort() {  
23        for (int i = 0; i < jmlData - 1; i++) {  
24            int idMin = i;  
25            for (int j = i + 1; j < jmlData; j++) {  
26                if (data[j] < data[idMin]) {  
27                    idMin = j;  
28                }  
29            }  
30            int temp = data[idMin];  
31            data[idMin] = data[i];  
32            data[i] = temp;  
33        }  
34    }
```

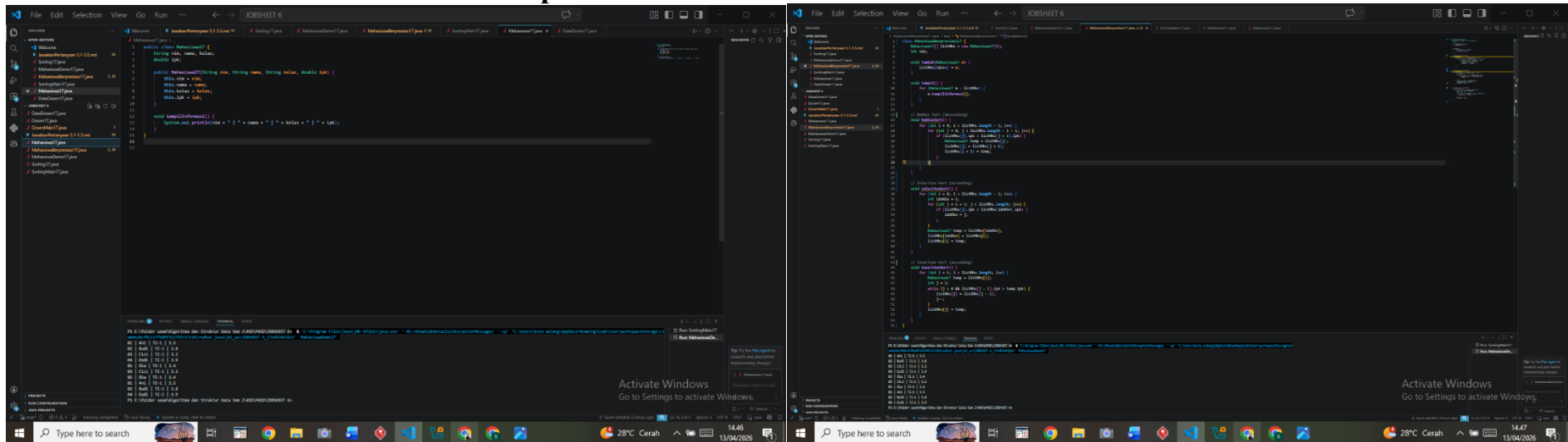
C. Insertion Sort

```
33     }
34 }
35
36 void InsertionSort() {
37     for (int i = 1; i < jmlData; i++) {
38         int temp = data[i];
39         int j = i;
40         while (j > 0 && data[j - 1] > temp) {
41             data[j] = data[j - 1];
42             j--;
43         }
44         data[j] = temp;
45     }
46 }
47
48 void tampil() {
```

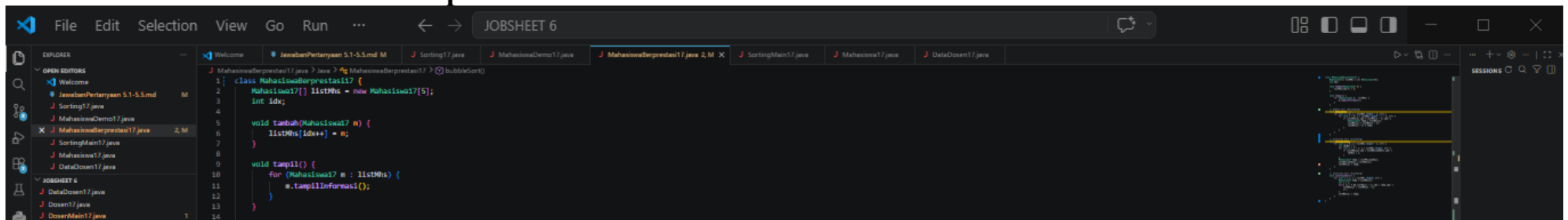
Class Sorting, Sorting Main & Hasil Running Program



5.3 Praktikum 2 (Array of Object) Class Mahasiswa & Class MahasiswaBerprestasi



Methode Tambah & Methode Tampil



Methode Bubble Sort, Methode Selection Sort & Insert Sort

```
14
15 // Bubble Sort (descending)
16 void bubbleSort() {
17     for (int i = 0; i < listDhs.length - 1; i++) {
18         for (int j = 0; j < listDhs.length - 1 - i; j++) {
19             if (listDhs[j].ipk < listDhs[j + 1].ipk) {
20                 Mahasiswa17 temp = listDhs[j];
21                 listDhs[j] = listDhs[j + 1];
22                 listDhs[j + 1] = temp;
23             }
24         }
25     }
26 }
27
28 // Selection Sort (ascending)
29 void selectionSort() {
30     for (int i = 0; i < listDhs.length - 1; i++) {
31         int idMin = i;
32         for (int j = i + 1; j < listDhs.length; j++) {
33             if (listDhs[j].ipk < listDhs[idMin].ipk) {
34                 idMin = j;
35             }
36         }
37         Mahasiswa17 temp = listDhs[idMin];
38         listDhs[idMin] = listDhs[i];
39         listDhs[i] = temp;
40     }
41 }
42
43 // Insertion Sort (ascending)
44 void insertionSort() {
45     for (int i = 1; i < listDhs.length; i++) {
46         Mahasiswa17 temp = listDhs[i];
47         int j = i - 1;
48         while (j > 0 && listDhs[j - 1].ipk > temp.ipk) {
49             listDhs[j] = listDhs[j - 1];
50             j--;
51         }
52         listDhs[j] = temp;
53     }
54 }
55 }
```

5.4 Insertion Sort

```
1 public class MahasiswaDemo17 {
2     public static void main(String[] args) {
3         MahasiswaBerprestasi17 mb = new MahasiswaBerprestasi17();
4         mb.tambah(new Mahasiswa17(nim: "01", nama: "Adi", kelas: "TI-1", ipk: 3.5));
5         mb.tambah(new Mahasiswa17(nim: "02", nama: "Budi", kelas: "TI-1", ipk: 3.8));
6         mb.tambah(new Mahasiswa17(nim: "03", nama: "Cici", kelas: "TI-1", ipk: 3.2));
7         mb.tambah(new Mahasiswa17(nim: "04", nama: "Dodi", kelas: "TI-1", ipk: 3.9));
8         mb.tambah(new Mahasiswa17(nim: "05", nama: "Eka", kelas: "TI-1", ipk: 3.4));
9
10        mb.tampil();
11        mb.insertionSort();
12        mb.tampil();
13    }
14 }
```

PS E:\Folder saun\Algoritma dan Struktur Data Sem 2\ASD\PA50\JOBSHEET 6 > "C:\Program Files\Java\jdk-19\bin\java.exe" -XX:+ShowCodeDetailsInExceptionMessages "-cp" "C:\U" @Run: SortingMain17

```
01 | Adi | TI-1 | 3,5
02 | Budi | TI-1 | 3,8
03 | Cici | TI-1 | 3,2
04 | Dodi | TI-1 | 3,9
05 | Eka | TI-1 | 3,4
06 | Cici | TI-1 | 3,2
07 | Eka | TI-1 | 3,4
08 | Adi | TI-1 | 3,5
09 | Budi | TI-1 | 3,8
10 | Dodi | TI-1 | 3,9
```

PS E:\Folder saun\Algoritma dan Struktur Data Sem 2\ASD\PA50\JOBSHEET 6 >

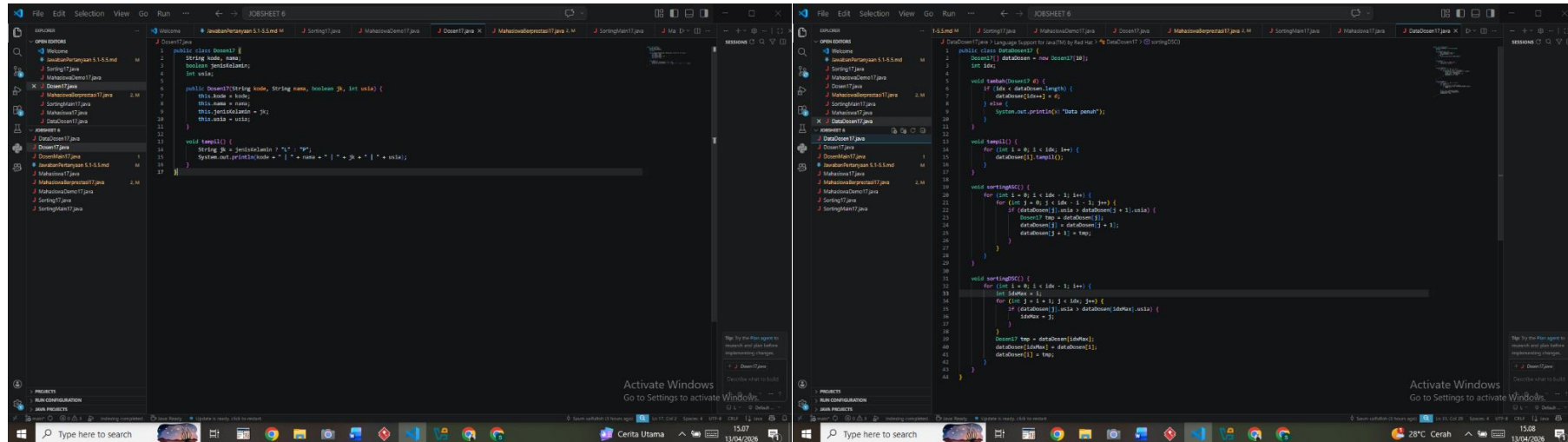
5.5 Latihan Praktikum (WAJIB)

Program Dosen

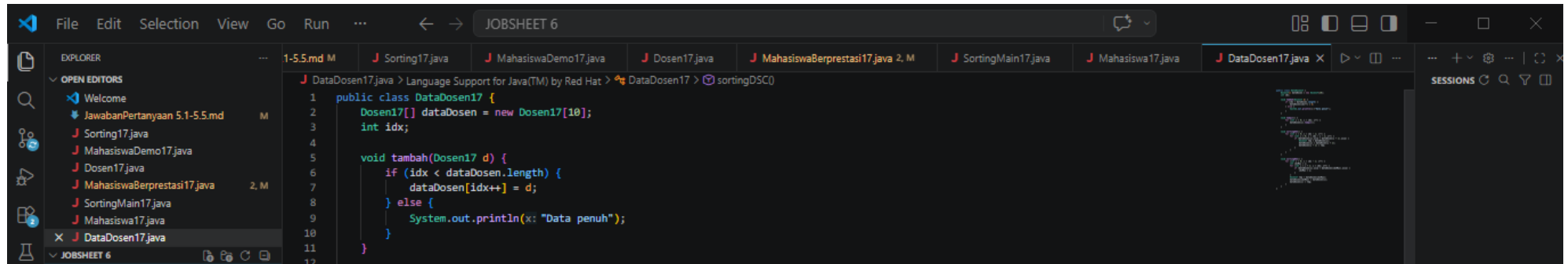
Fitur

1. Tambah Data
2. Tampil Data
3. Sorting ASC (Bubble Sort)
4. Sorting DSC (Selection / Insertion)

Class Dosen & Class Data Dosen



Metode Tambah



Methode SortingASC

```
J Mahasiswa17.java
J MahasiswaBerprestasi17.java 2, M
J MahasiswaDemo17.java
J Sorting17.java
J SortingMain17.java

17 }
18
19 void sortingASC() {
20     for (int i = 0; i < idx - 1; i++) {
21         for (int j = 0; j < idx - i - 1; j++) {
22             if (dataDosen[j].usia > dataDosen[j + 1].usia) {
23                 Dosen17 tmp = dataDosen[j];
24                 dataDosen[j] = dataDosen[j + 1];
25                 dataDosen[j + 1] = tmp;
26             }
27         }
28     }
29 }
30 }
```

Methode SortingDSC

```
29 }
30
31 void sortingDSC() {
32     for (int i = 0; i < idx - 1; i++) {
33         int idxMax = i;
34         for (int j = i + 1; j < idx; j++) {
35             if (dataDosen[j].usia > dataDosen[idxMax].usia) {
36                 idxMax = j;
37             }
38         }
39         Dosen17 tmp = dataDosen[idxMax];
40         dataDosen[idxMax] = dataDosen[i];
41         dataDosen[i] = tmp;
42     }
43 }
44 }
```

Tip: Try the [Plan agent](#) to research and plan before implementing changes.

Describe what to build

Activate Windows
Go to Settings to activate Windows.

Saum saifulloh (3 hours ago) Ln 33, Col 28 Spaces: 4 UTF-8 CRLF Java

15.08
28°C Cerah
13/04/2026

Menu Program Main

```
J Mahasiswa17.java
J DataDosen17.java
X J DosenMain17.java 2
JOB SHEET 6
J DataDosen17.java
J Dosen17.java
J DosenMain17.java 2
J JawabanPertanyaan 5.1-5.5.md M
J Mahasiswa17.java
J MahasiswaBerprestasi17.java 2, M
J MahasiswaDemo17.java

7 int pilih;
8
9 do {
10     System.out.println("MENU");
11     System.out.println("1. Tambah Data Dosen");
12     System.out.println("2. Tampil Data Dosen");
13     System.out.println("3. Sorting ASC (Usia Termuda -> Tertua)");
14     System.out.println("4. Sorting DSC (Usia Tertua -> Termuda)");
15     System.out.println("0. Keluar");
16     System.out.print("Pilih: ");
17     pilih = sc.nextInt(); sc.nextLine();
18 }
```

Input Data

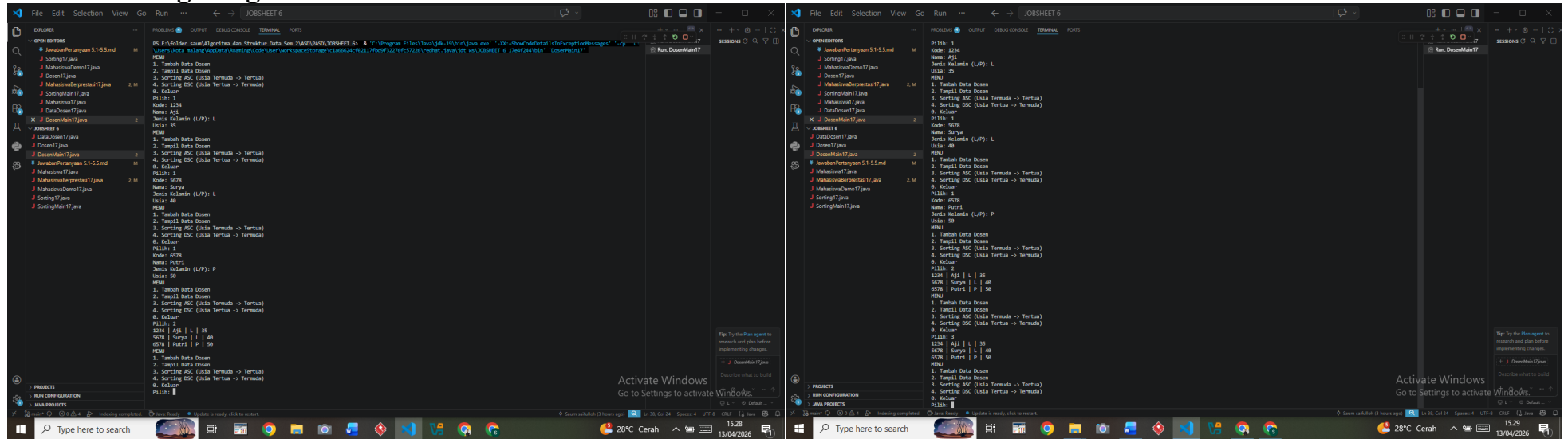
The screenshot displays an IDE with the following components:

- EXPLORER:** Lists files including `JawabanPertanyaan 5.1-5.5.md`, `Sorting17.java`, `MahasiswaDemo17.java`, `Dosen17.java`, `MahasiswaBerprestasi17.java`, `SortingMain17.java`, `Mahasiswa17.java`, `DataDosen17.java`, and `DosenMain17.java`.
- EDITOR:** Shows the code for `DosenMain17.java`. The code is as follows:

```
1 import java.util.Scanner;
2
3 public class DosenMain17 {
4     Run | Debug | Run main | Debug main
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         DataDosen17 dd = new DataDosen17();
8         int pilih;
9
10        do {
11            System.out.println("MENU");
12            System.out.println("1. Tambah Data Dosen");
13            System.out.println("2. Tampil Data Dosen");
14            System.out.println("3. Sorting ASC (Usia Termuda -> Tertua)");
15            System.out.println("4. Sorting DSC (Usia Tertua -> Termuda)");
16            System.out.println("0. Keluar");
17            System.out.print("Pilih: ");
18            pilih = sc.nextInt(); sc.nextLine();
19
20            switch (pilih) {
21                case 1:
22                    System.out.print("Kode: ");
23                    String kd = sc.nextLine();
24                    System.out.print("Nama: ");
25                    String nm = sc.nextLine();
26                    System.out.print("Jenis Kelamin (L/P): ");
27                    boolean jk = sc.nextLine().equalsIgnoreCase("L");
28                    System.out.print("Usia: ");
29                    int us = sc.nextInt(); sc.nextLine();
30                    dd.tambah(new Dosen17(kd, nm, jk, us));
31                    break;
32                case 2:
33                    dd.tampil();
34            }
35        } while (pilih != 0);
36    }
37 }
```
- TERMINAL:** Shows the command to run the application and its output:

```
PS E:\folder saum\Algoritma dan Struktur Data Sem 2\ASD\PASD\JOBSHEET 6> & 'C:\Program Files\Java\jdk-19\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\kota_malang\AppData\Roaming\Code\User\workspaceStorage\c1a66624cf82117fbd9f32276fc57226\redhat.java\jdt_ws\JOBSHEET_6_17e4f244\bin' 'DosenMain17'
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Usia Termuda -> Tertua)
4. Sorting DSC (Usia Tertua -> Termuda)
0. Keluar
Pilih:
```
- STATUS BAR:** Shows the current file is `main*`, the IDE is ready, and the Java version is `Java: Ready`.

Hasil Running Program



The image displays two screenshots of an IDE (IntelliJ IDEA) showing the execution of a Java program. The program is titled "JOBSHEET 6" and is located in the file "J:\DataDosen17.java". The program's output is displayed in the "OUTPUT" window, showing the results of the sorting process.

Left Screenshot: The program is running, and the output window shows the initial state of the data. The output is as follows:

```
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 1
Kode: 1234
Nama: Suriya
Jenis Kelamin (L/P): L
Usia: 35
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 1
Kode: 6578
Nama: Putri
Jenis Kelamin (L/P): P
Usia: 50
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 1
Kode: 1234
Nama: Suriya
Jenis Kelamin (L/P): L
Usia: 35
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 1
Kode: 6578
Nama: Putri
Jenis Kelamin (L/P): P
Usia: 50
MENU
```

Right Screenshot: The program is running, and the output window shows the results of the sorting process. The output is as follows:

```
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 1
Kode: 1234
Nama: Suriya
Jenis Kelamin (L/P): L
Usia: 35
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 1
Kode: 6578
Nama: Putri
Jenis Kelamin (L/P): P
Usia: 50
MENU
1234 | A | L | 35
6578 | S | L | 40
6578 | P | P | 50
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 2
1234 | A | L | 35
6578 | S | L | 40
6578 | P | P | 50
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 3
1234 | A | L | 35
6578 | S | L | 40
6578 | P | P | 50
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 4
6578 | P | P | 50
6578 | S | L | 40
1234 | A | L | 35
MENU
1. Tambah Data Dosen
2. Tampil Data Dosen
3. Sorting ASC (Ukura Tertua -> Tertua)
4. Sorting DSC (Ukura Tertua -> Tertua)
0. Keluar
Pilih: 0
```

Link GitHub: <https://github.com/saumsaifulloh/AlgoritmaDanStrukturDataSem2/tree/main/ASD/PASD/JOBSHEET%206>

